JOSE GARCIA

https://jgarc521.github.io
San Luis Obispo, CA 93405
(408) 593-6974 ⋄ jgarc.0904@gmail.com

EDUCATION

Master of Science in Statistics Expected June 2026
California Polytechnic State University, San Luis Obispo

Bachelor of Science in Statistics June 2025
California Polytechnic State University, San Luis Obispo

PROFESSIONAL EXPERIENCE

Data Scientist (Graduate Researcher) — Ecological Time-Series Analytics Oct 2024 – Present
Cal Poly, San Luis Obispo

- Designed and implemented multivariate time-series pipelines in R to analyze 100,000+ high-frequency sensor readings for environmental monitoring.
- Built production-style QA/QC workflows using QARTOD rules, reducing data anomalies by 30% and standardizing hourly aggregation.
- Developed forecasting and anomaly-detection models using state-space methods and Kalman filtering to identify trends and seasonal patterns in estuary conditions.
- Communicated model insights to faculty and research partners, informing decisions on sensor maintenance and estuarine monitoring strategies.

Data Analyst — Housing Insecurity Modeling Jan 2025 – Mar 2025
Cal Poly, San Luis Obispo

- Built logistic regression models to identify key economic and demographic drivers of housing insecurity using county-level administrative data.
- Conducted feature engineering, cross-validation, and diagnostic evaluation, improving model accuracy by 12%.
- Delivered clear, actionable summaries to community partners, highlighting high-risk groups and data-driven opportunities for resource allocation.

Environmental Monitoring Intern Sep 2023 – Jun 2024
Morro Bay National Estuary Program *Morro Bay, CA*

- Cleaned, analyzed, and visualized multi-year CO_2 sensor data to support air-quality and estuarine research.
- Automated data cleaning and reporting workflows in R, reducing processing time by 40%.
- Produced dashboards and technical reports used by program staff and partner agencies.

TECHNICAL SKILLS

Programming	R (tidyverse, tidymodels, MARSS), Python, SQL, Julia
Data Science	Predictive modeling, time-series forecasting, ML models, feature engineering
ML/Stats	State-space models, GLMs, PCA, clustering, SVM, random forests, cross-validation

PROJECTS

BRFSS Alcohol Use Classification Fall 2025
R, tidymodels, glmnet, ranger

- Built classification models (logistic regression, LASSO, SVM, random forest) to predict binge drinking behavior from 10,000+ BRFSS survey records.
- Preprocessed survey data, managed missingness, created features, and ran 5-fold cross-validation; compared model ROC-AUC to select final model.
- Visualized key predictors and generated stakeholder-ready summaries for non-technical audiences.